# Ownership Structure and Economic Growth: A Network Approach

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# Ownership Structure $\Longrightarrow$ Economic Growth?

- The rise of common ownership
  - BlackRock, Vanguard, and State Street control a quarter of votes of S&P 500 firms
  - Firms maximize shareholder values $\Longrightarrow$
    Partially internalize externalities for commonly owned firms
- Ownership structure (common ownership / cross ownership / M&A) $\rightarrow$ economic growth?
  - Business stealing
  - Technology spillovers

# Framework: Growth + IO + Ownership

- Quantitative endogenous growth (Hopenhayn and Okumura, 2024) with ownership structure
- Dynamic R&D involving many oligopolists
  - Existing models: a few firms in each industry (mostly monopolistic competition or duopoly)
  - Our exercise: Public patenting firms in the US ($\simeq$ 1000 firms)
- Quantify three inter-firm networks
  - Ownership structure
  - Product market rivalry
  - Technology spillovers

# Quantify the Impact of Common Ownership and Policy

- Concentration of ownership structure affecting R&D decisions:
  - Internalization of business stealing $\Longrightarrow g \downarrow$
  - Internalization of technology spillover $\Longrightarrow g \uparrow$
  - Dispersed ownership $\Longrightarrow$ Observed common ownership: $g \downarrow$ by 0.084 pp
- Uniform R&D subsidy
  - Optimal rate is $s = 33\%$
  - $g$ increases by 0.25 pp
  - CE Welfare loss relative to optimal R&D allocation is reduced to 0.1% (Initially 0.7%)

# Literature

- Competition & innovation:
  d'Aspremont and Jacquemin (1988); Aghion et al. (2001, 2005); Acemoglu and Akcigit (2012); Aghion et al. (2013); Bloom et al. (2013); Lopez and Vives (2019); Peters (2020); Akcigit and Ates (2021, 2023); Liu et al. (2022); Cavenaile et al. (2023); Anton et al. (2023, 2024); Kini et al. (2024)
  Quantitative endogenous growth model matched to firm-(pair-)level data
  Ownership structure $\Longrightarrow$ economic growth
- Hedonic demand / empirical IO:
  Lancaster (1966); Rosen (1974); Epple (1987); Berry et al. (1995); Nevo (2001); Pellegrino (2024); Ederer and Pellegrino (2024)
  Dynamic general equilibrium / R&D
- Oligopoly / common ownership / market power:
  Rubinstein and Yaari (1983); Rotemberg (1984); Neary (2003); Atkeson and Burstein (2008); Gutierrez and Philippon (2017); He and Huang (2017); Azar et al. (2018, 2022); Autor et al. (2020); Baqaee and Farhi (2020); De Loecker et al. (2020); Azar and Vives (2021); Edmond et al. (2023)

# Overview of Framework and Identification

- Firms compete to maximize shareholders' returns $\Longrightarrow$
  - Firms partially consider the values of other firms owned by the same shareholders
- Overlapping ownership among firms that are close in ...
  - product space $\Longrightarrow$ internalize business stealing effect $\Longrightarrow$ R&D $\downarrow$
  - technology space $\Longrightarrow$ internalize technology spillover $\Longrightarrow$ R&D $\uparrow$

# Overview of Framework and Identification

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  - technology space $\Longrightarrow$ internalize technology spillover $\Longrightarrow$ R&D $\uparrow$

| Networks | Measurement |
|---|---|
| Ownership structure | Investor holdings in 13f filings (Backus et al., 2021) |
| Product market rivalry | Product proximity (Hoberg and Phillips, 2016):<br>Text analysis of product description in 10k filings |
| Technology spillover | Technology proximity (Bloom et al., 2013):<br>Patent classification |

# Generalized Hedonic-Linear Demand (Pellegrino, 2024) GHL

- $i \in \{1, 2, ..., n\}$: firms / products
- $\mathbf{q}_t = [q_1, q_2, ...q_n]$: vector of quantities
- Linear quadratic aggregator:

$$Y_t = \mathbf{q}_t^T \mathbf{b}_t - \frac{1}{2}\mathbf{q}_t^T \Sigma \mathbf{q}_t$$

- Linear inverse demand:

$$\mathbf{p}_t = \mathbf{b}_t - \Sigma \mathbf{q}_t$$

- $\mathbf{b}_t$: vector of product qualities
- $\Sigma$: matrix of product substitutability

# Production Technology and Technology Choice

- Production technology: $q_{i,t} = a_{i,t} l_{i,t}$
- Each firm has their knowledge capital $z_{i,t}$
- Each firm allocates their knowledge capital to improve product quality and labor productivity:

$$\zeta a_{i,t} + b_{i,t} = z_{i,t}$$

# Law of Motion of Knowledge Capital

$$d\mathbf{z}_t = \left( \underbrace{\Omega \mathbf{z}_t}_{\text{technology spillover}} + \underbrace{\mu \mathbf{x}_t}_{\text{R\&D}} \right) dt + \underbrace{\gamma \mathbf{z}_t dW_t}_{\text{shocks}}$$

- $\Omega = [\omega_{ij}]$: technology spillover matrix
- $x_{i,t} = \sqrt{d_{i,t}}$
  - $d_{i,t}$: R&D cost in terms of final good
  - Innovation elasticity is 0.5
- $\mu$, $\gamma$: positive scalars

# Market Clearing and Preference

- Inelastic labor supply:

$$L = \sum_i l_{i,t}$$

- Final good market clearing:

$$C_t + \underbrace{\sum_i d_{i,t}}_{\text{R\&D cost}} = Y_t$$

- Risk neutral representative household:

$$\max \mathbb{E}_t \left[ \int_t^\infty \exp\left(-\rho s\right) C_s ds \right]$$

# Ownership Structure

- $z \in \{1, 2, \ldots, n_z\}$: funds
- $s_{iz}$: the proportion of shares in firm $i$ owned by fund $z$ where $\sum_z s_{iz} = 1$
- $\widehat{V}_{i,t}$: value of firm $i$
- $\widetilde{V}_{z,t} \equiv \sum_i s_{iz} \widehat{V}_{i,t}$: value of fund $z$

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- Firms' objective:

$$\sum_z s_{iz} \widetilde{V}_{z,t} \propto \sum_j \kappa_{ij} \widehat{V}_{j,t}$$

- $K = [\kappa_{ij}]$: common ownership weights (Rotemberg, 1984; Ederer and Pellegrino, 2024)
  - All diagonal elements are 1
  - All off-diagonal elements are 0 $\Longleftrightarrow$ dispersed ownership: each firm maximizes their value
  - All elements are 1 $\Longleftrightarrow$ monopoly: maximizes producer surplus

# Cournot-Nash Equilibrium

- Firm $i$'s gross profit before subtracting dynamic R&D cost:

$$\pi_{i,t} = p_{i,t} q_{i,t} - w_t l_{i,t} = \left( b_{i,t} - \frac{1}{a_{i,t}} w_t - \sum_j \sigma_{ij} q_{j,t} \right) q_{i,t} \quad \text{where} \quad \zeta a_{i,t} + b_{i,t} = z_{i,t}$$

- Given $w_t$, $z_{i,t}$, and $\{q_{j,t}\}_{j \neq i}$, firm $i$ chooses $a_{i,t}$, $b_{i,t}$, and $q_{i,t}$ to maximize $\sum_j \kappa_{ij} \pi_{j,t}$

# Cournot-Nash Equilibrium

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- Quantity is a linear function of knowledge capital:

$$\mathbf{q}_t = \left\{ \underbrace{2\frac{\zeta}{L} J}_{\text{labor cost}} + \underbrace{\Sigma}_{\text{substitutability}} + \underbrace{K \circ \Sigma}_{\text{ownership weight} \times \text{substitutability}} \right\}^{-1} \mathbf{z}_t$$

  - $J$: matrix with all elements equal to 1
- Ownership-weighted gross profits are expressed in quadratic form: $\sum_j \kappa_{ij} \pi_{j,t} = \mathbf{z}_t^T Q^i \mathbf{z}_t$ Q

# Linear-Quadratic Differential Game

- Given other players' strategy $\{x_{j,t}\}_{j\neq i, t\geq 0}$, firm $i$ chooses R&D effort $\{x_{i,t}\}_{t\geq 0}$ to maximize

$$\max_{\{x_{i,t}\}_{t\geq 0}} \quad V^i(\mathbf{z}_0) \equiv \mathbb{E}_0\left[\int_0^\infty \exp(-\rho t)\left\{\sum_j \kappa_{ij}\left(\pi_{j,t} - x_{j,t}^2\right)\right\} dt\right]$$

subject to $d\mathbf{z}_t = (\Omega\mathbf{z}_t + \mu\mathbf{x}_t)\,dt + \gamma\mathbf{z}_t dW_t$

- Firm $i$'s HJB equation:

$$\rho V^i(\mathbf{z}) = \max_{x_i}\left\{\mathbf{z}^T Q^i \mathbf{z} - \sum_j \kappa_{ij} x_j^2 + V_{\mathbf{z}}^i(\mathbf{z})\left[\Omega\mathbf{z} + \mu\mathbf{x}\right] + \frac{\gamma^2}{2}\mathbf{z}^T V_{\mathbf{zz}}^i(\mathbf{z})\mathbf{z}\right\}$$

# HJB Equations $\Longrightarrow$ Riccati Equations

- Guess and verify $V^i(\mathbf{z}) = \mathbf{z}^T X^i \mathbf{z}$ (for any $\mathbf{z}$)
- $X^i$ is the solution of stacked algebraic Riccati equations Riccati
- All public patenting firms in the US in our dataset $\simeq$1000 firms $\Longrightarrow$
  - $1000^3 = 1$ billion undetermined coefficients

| Oligopolistic Schumpeterian growth model | Computation time ($n$ firms) | Number of firms | Space of productivity |
|---|---|---|---|
| Cavenaile et al. (2023) | $O(2^n)$ | 4 | 6 grids |
| Our model | $O(n^4)$ | Around 1000 | Continuous |

# BGP Equilibrium

- R&D strategy: $x_{i,t} = \left(\mu X_i^i\right)^T \mathbf{z}_t$ where $X_i^i$ is the $i$ th column of $X^i$
- The law of motion is rewritten as $d\mathbf{z}_t = \Phi \mathbf{z}_t dt + \gamma \mathbf{z}_t dW_t$ where

$$\underbrace{\Phi}_{\text{technology transition matrix}} \equiv \underbrace{\Omega}_{\text{spillover}} + \underbrace{\mu^2 \begin{bmatrix} X_1^1 & \cdots & X_n^n \end{bmatrix}^T}_{\text{R\&D}}$$

## Theorem

*Consider the deterministic economy ($\gamma = 0$). If the technology transition matrix $\Phi$ is irreducible,*
*(i) there exists a largest positive eigenvalue of $\Phi$, $g$, and associated strictly positive eigenvector, $\mathbf{z}^*$*
*(ii) there exists a globally stable BGP such that the knowledge capital growth rate of all firms is $g$ and knowledge capital distribution is a scalar multiple of $\mathbf{z}^*$*

- Proof: Perron–Frobenius Theorem
- "$\Phi$ is irreducible" $\Longleftrightarrow$ "All firms are directly or indirectly connected technologically"

# Aggregation

- Output is expressed in quadratic form: Equilibrium Summary

$$Y_t = \mathbf{q}_t^T \mathbf{b}_t - \frac{1}{2}\mathbf{q}_t^T \Sigma \mathbf{q}_t = \mathbf{z}_t^T Q \mathbf{z}_t$$

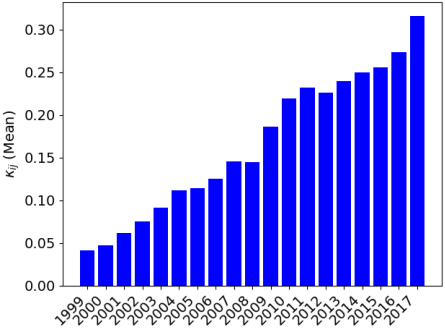
  - $\mathbf{a}_t$, $\mathbf{b}_t$, $\mathbf{z}_t$ , and $\mathbf{q}_t$ grow at the same rate on BGP in deterministic economy ($\gamma = 0$).
- Apply Ito's lemma to $\log Y_t = \log\left(\mathbf{z}_t^T Q \mathbf{z}_t\right)$ and $d\mathbf{z}_t = \Phi \mathbf{z}_t dt + \gamma \mathbf{z}_t dW_t \Longrightarrow g_t$ Decomposition
- Expected utility is expressed in quadratic form: X

$$\mathbb{E}_t\left[\int_t^{\infty} \exp\left(-\rho s\right) C_s ds \middle| \mathbf{z}_t\right] = \mathbf{z}_t^T X \mathbf{z}_t$$

- Solve the equilibrium once $\Longrightarrow$ Can compute expected growth and utility for any $\mathbf{z}_t$

# Common Ownership Weight

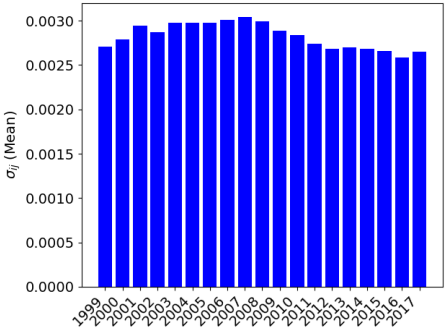
- Form 13F: Required for all institutional investment managers with at least $100 million in assets
- Backus et al. (2021) construct a dataset on investors' holdings based on Form 13F
- Common ownership weight $\kappa_{ij} \equiv \mathbf{s}_i^T \mathbf{s}_j / \mathbf{s}_i^T \mathbf{s}_i$ where $\mathbf{s}_i \equiv [s_{i1}, \ldots, s_{iz}, \ldots, s_{in_z}]^T$



# Product Substitutability

$$\underbrace{\sigma_{ij}}_{\text{substitutabiliy}} = \alpha \times \text{product similarity b/w } i \text{ and } j \quad (i \neq j)$$

- Hoberg and Phillips (2016) estimates product similarity using product descriptions in 10-K
- Pellegrino (2024) estimates $\alpha$ to align with the cross-price elasticity of demand



# Technological Proximity

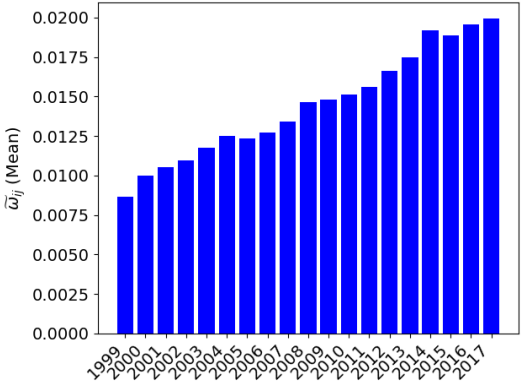
- $T_i$: the vector representing the share of patents held by firm $i$ in each technology class
  - (i) subclass-level ($\approx$ 600) or (ii) group-level ($\approx$ 4000) classifications
- Jaffe (1986) technology proximity measure:

$$\tilde{\omega}_{ij} = \frac{T_i T_j'}{\left(T_i T_i'\right)^{1/2} \left(T_j T_j'\right)^{1/2}}$$

  - Assumption: Spillovers exist only within the same technology classes
- Mahalanobis measure for robustness (Bloom et al., 2013; Arora et al., 2021)
  - Accounts for spillovers between technology classes
  - Based on frequency with which patents are taken out in different classes within firm

# Technological Proximity

- Merge USPTO data with Compustat firms using DISCERN 2 dataset (Arora et al., 2024)
- Jaffe measure, Group-level patent classification, Stack for 5 years



# Initial Distribution of Knowledge Capital

1. Profit $\pi_{i,t} \Longleftarrow$ "Revenue" $-$ "Costs of goods sold" in Compustat
   - Deflated by GDP deflator
2. Quantity $\mathbf{q}_t \Longleftarrow \pi_{i,t} = \sum_j \kappa_{ij}\sigma_{ij}q_{i,t}q_{j,t}$
3. $\zeta/L \Longleftarrow$ match the total cost of the sample firms
4. Knowledge capital $\mathbf{z}_t \Longleftarrow \mathbf{q}_t = \left\{2\frac{\zeta}{L}J + \Sigma + K \circ \Sigma\right\}^{-1}\mathbf{z}_t$

# Technology Spillover $\Omega = \beta \times$ Technology Proximity $\tilde{\Omega}$

$$z_{i,t+1} - z_{i,t} = \beta \sum_{j \neq i} \tilde{\omega}_{ij,t} z_{j,t} + \text{Year FE}_t + \varepsilon_{i,t}$$

| | (1) | (2) | (3) |
|---|---|---|---|
| | $z_{i,t+1} - z_{i,t}$ | $z_{i,t+1} - z_{i,t}$ | $z_{i,t+1} - z_{i,t}$ |
| $\sum_{j \neq i} \tilde{\omega}_{ij,t} z_{j,t}$ | 0.000191*** | 0.0000152*** | 0.000140*** |
| | (0.000035) | (0.000035) | (0.000039) |
| $\sqrt{\text{R\&D Expenditure}}$ | | .037** | |
| | | (0.021) | |
| Year Fixed Effects | ✓ | ✓ | ✓ |
| IV | | | ✓ |
| IV 1st Stage F-statistics | | | 4176 |
| Patent Classification | Group | Group | Group |
| Technological Proximity | Jaffe | Jaffe | Jaffe |
| No. observations | 16,324 | 15,173 | 14,181 |

SEs clustered by years and 4-digit naics industries are reported in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. First Stage

- IV: User cost of R&D, driven by state-level tax variations (Wilson, 2009; Bloom et al., 2013)

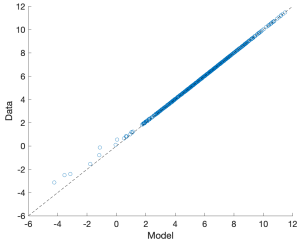
# Identification: Summary

- Publicly available data + Compustat

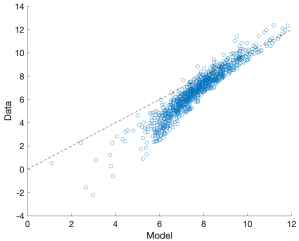
| Notation | Description | Value | Source |
|---|---|---|---|
| $\Psi^T\Psi$ | Product proximity | | Form 10-K, Hoberg and Phillips (2016) |
| $\tilde{\Omega}$ | Technological proximity | | USPTO, Patent classification |
| $K$ | Common ownership weights | | Form 13F, Backus et al. (2021) |
| $\alpha$ | Product proximity $\rightarrow$ Substitutability | .12 | Pellegrino (2024) |
| $\beta$ | Technological proximity $\rightarrow$ Spillover | .00014 | Estimate the law of motion |
| $\gamma$ | St.d. of idiosyncratic shocks | .027 | Estimate the law of motion |
| $\zeta/L$ | Labor augmentation efficiency | .0063 | Compustat, Cost of goods sold |
| $\rho$ | Discount rate | .10 | |
| $\mu$ | R&D efficiency | .05 | 1.6% economic growth rate |

# Goodness of Fit b/w Model and Data

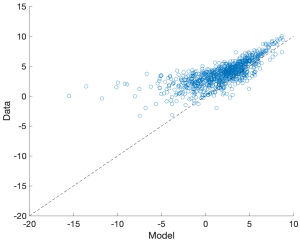
- Model ($x$-axis) vs Data ($y$-axis) : firm-level profits (targeted), sales and R&D expenditures




(a) Log of Profit (Targeted)

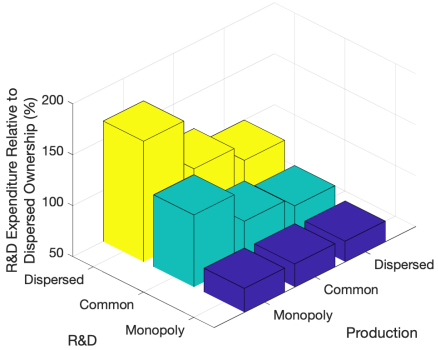



(b) Log of Sales

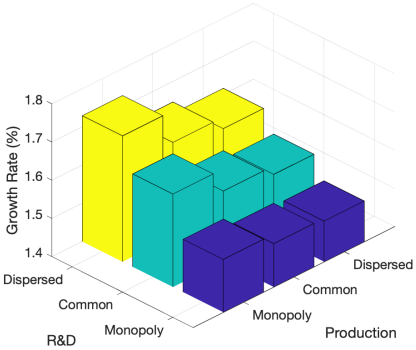



(c) Log of R&D Expenditure

# Different Ownership Structure: R&D Expenditure and Growth Rate



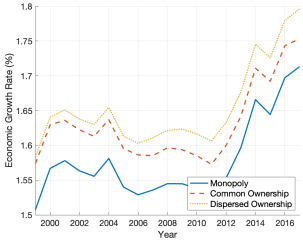

(a) R&D Expenditure



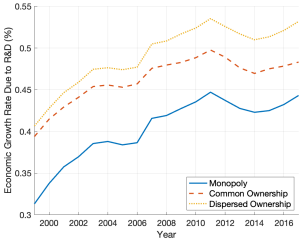

(b) Expected Growth Rate

- Concentration in production: Appropriability of return from innovation $\Longrightarrow g \uparrow$
- Concentration in R&D: internalization of business stealing $>$ technology spillover $\Longrightarrow g \downarrow$ Welfare

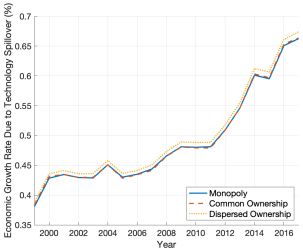
# Growth Rates under Common Ownership Approach Monopoly Level




(a) Expected Economic Growth Rates



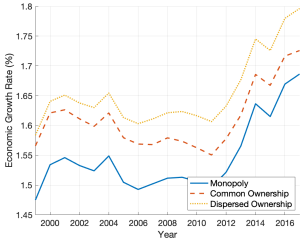

(b) Growth Due to R&D



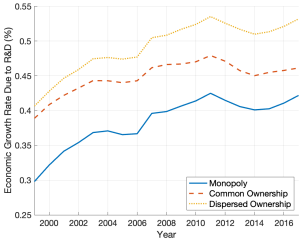

(c) Growth Due to Tech Spillover

- Economic growth rate under common ownership deviates from that under dispersed ownership and approaches that under monopoly
- Dispersed ownership $\Longrightarrow$ Observed common ownership: $g \downarrow$ by 0.043 pp in 2017

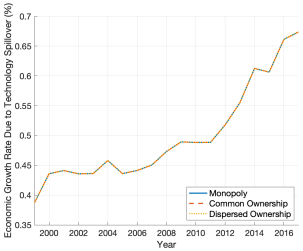
# Growth Rates (Only Ownership Structure of R&D Differs)




(a) Expected Economic Growth Rates




(b) Growth Due to R&D

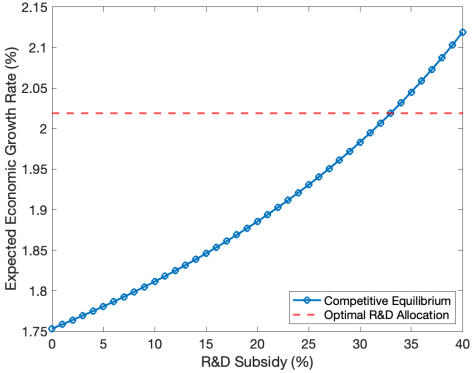



(c) Growth Due to Tech Spillover

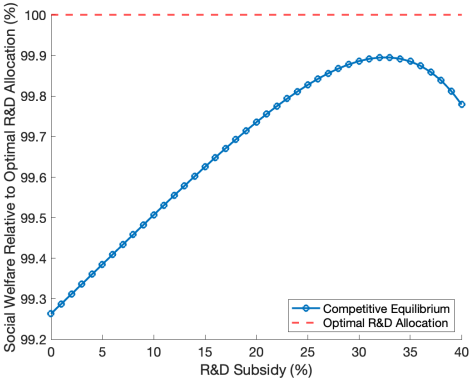
- Economic growth rate under common ownership deviates from that under dispersed ownership and approaches that under monopoly
- Dispersed ownership $\Longrightarrow$ Observed common ownership: $g \downarrow$ by 0.084 pp in 2017

# Uniform R&D Subsidy

Social Optimum




(a) Expected Growth



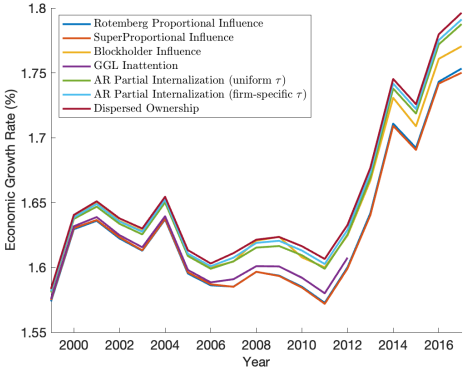

(b) Expected Social Welfare

- Optimal rate is $s = 33\%$, which increases $g$ by 0.25 pp
- CE Welfare loss relative to optimal R&D allocation is reduced to 0.1% (Initially 0.7%)

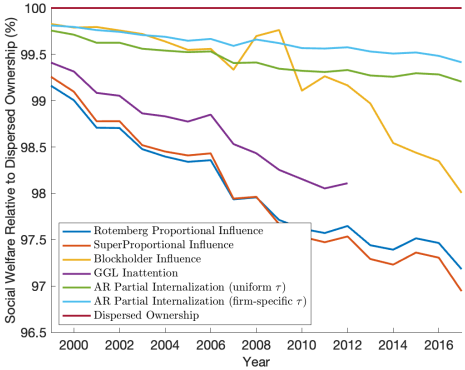
# Alternative Corporate Governance Models: Ederer and Pellegrino (2024)

1. Super-proportional influence: $\tilde{\kappa}_{ij} = \frac{\sum_{z=1}^{Z} s_{iz}\gamma_{iz}s_{jz}}{\sum_{z=1}^{Z} s_{iz}\gamma_{iz}s_{iz}}$ where $\gamma_{iz} = \sqrt{s_{iz}}$
2. Blockholder influence: $\tilde{\kappa}_{ij} = \frac{\sum_{z=1}^{Z} s_{iz}b_{iz}s_{jz}}{\sum_{z=1}^{Z} s_{iz}s_{jz}} \quad (i \neq j)$, where $b_{iz} = 1$ if $s_{iz} > 5\%$
3. Rational investor inattention
   - Gilje et al. (2020) (GGL) estimate the probability that an investor votes against Institutional Shareholders Service recommendations
   - Utilize the estimate to capture the investor's level of attention
4. Governance frictions and entrenchment
   - Azar and Ribeiro (2021) (AR) estimate an objective function where the manager of firm $i$ discounts other firms' profit by $\tau_i$

# Alternative Corporate Governance Models




(a) Expected Growth




(b) Expected Social Welfare

# Conclusion

- Quantitative endogenous growth model with ownership structure
  - Match to data at firm (pair) level
- Common ownership in R&D:
  - Internalization of business stealing $\Longrightarrow g \downarrow$
  - Internalization of technology spillover $\Longrightarrow g \uparrow$
  - Overall, the observed rise of common ownership decreases $g$
- Potential applications:
  - The impact of the dissolution of Zaibatsu in Japan after WWII
  - The relationship b/w Chaebols and South Korea's growth miracle

# Generalized Hedonic-Linear Demand (Pellegrino, 2024)

- $i \in \{1, 2, \ldots, n\}$: firms / products
- 1 unit of product $i$ provides
  - 1 unit of idiosyncratic characteristic $k \in \{1, 2, \ldots, n\}$
  - $\psi_{k,i}$ unit of shared characteristic $k \in \{n+1, n+2, \ldots, n+n_k\}$ where $\sum_k \psi_{k,i}^2 = 1$
- Aggregate each characteristic:

$$y_{k,t} = \begin{cases} q_{k,t} & k = 1, 2, \ldots, n \\ \sum_i \psi_{k,i} q_{i,t} & k = n+1, n+2, \ldots, n+n_k \end{cases}$$

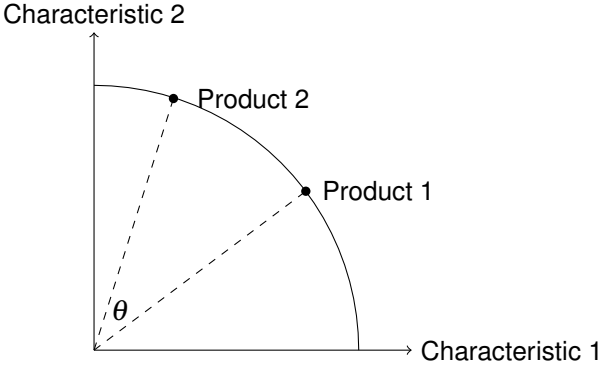
- Linear quadratic aggregator over characteristics:

$$Y_t = (1-\alpha) \sum_{k=1}^{n} \left( \underbrace{\hat{b}_{k,t} y_{k,t} - \frac{1}{2} y_{k,t}^2}_{\text{idiosyncratic characteristic}} \right) + \alpha \sum_{k=n+1}^{n+n_k} \left( \underbrace{\hat{b}_{k,t} y_{k,t} - \frac{1}{2} y_{k,t}^2}_{\text{shared characteristic}} \right)$$

# Product Substitutability $\propto$ Product Proximity (Pellegrino, 2024)

- $\Sigma = [\sigma_{ij}]$ is estimated using product proximity measure
- Product proximity is based on text analysis of product description in 10k filings

$$\sigma_{ij} = \alpha \times \text{product cosine similarity b/w } i \text{ and } j \quad (i \neq j)$$



# Generalized Hedonic-Linear Demand (Pellegrino, 2024)

- Quality:

$$b_i = (1-\alpha)\,\hat{b}_i + \alpha \sum_{k=n+1}^{n+n_k} \psi_k \hat{b}_k$$

- Inverse demand:

$$\frac{\mathbf{p}}{P} = \mathbf{b} - \Sigma \mathbf{q}$$

- Inverse cross price elasticity of demand:

$$\frac{\partial \log p_i}{\partial \log q_j} = -\frac{q_j}{p_i} \cdot \sigma_{ij}$$

- Cross price elasticity of demand:

$$\frac{\partial \log q_i}{\partial \log p_j} = -\frac{p_j}{q_i} (\Sigma^{-1})_{ij}$$

Back

# Static Profits

- Real gross profit: $\frac{\pi_{i,t}}{P_t} = \sum_j \kappa_{ij}\sigma_{ij}q_{i,t}q_{j,t}$
- Firms choose labor productivity and product quality: $\zeta a_{i,t} = \sqrt{\zeta \frac{w_t}{P_t}}$, $b_{i,t} = z_{i,t} - \sqrt{\zeta \frac{w_t}{P_t}}$
- Labor market clearing: $L = \sum_i \frac{q_{i,t}}{a_{i,t}} \Longrightarrow \sqrt{\zeta \frac{w_t}{P_t}} = \frac{\zeta}{L}\sum_i q_{i,t}$
- $\mathbf{q}_t = N\mathbf{z}_t$ where $N \equiv \left\{2\frac{\zeta}{L}J + \Sigma + K \circ \Sigma\right\}^{-1}$
- $N_i$: the $i$ th row of $N$
- Ownership weighted profit:

$$\sum_j \kappa_{ij}\frac{\pi_{j,t}}{P_t} = \sum_j \kappa_{ij}\sum_h \kappa_{jh}\sigma_{jh}q_{j,t}q_{h,t} = \mathbf{z}_t^T Q^i \mathbf{z}_t$$

where

$$Q^i = \frac{1}{2}\sum_j \kappa_{ij}\sum_h \kappa_{jh}\sigma_{jh}\left(N_j^T N_h + N_h^T N_j\right)$$

Back

# Riccati Equations

- $V^i(\mathbf{z}) = \mathbf{z}^T X^i \mathbf{z}$ where $X^i$ is the solution of the stacked Riccati equation

$$0 = Q^i - \mu^2 \sum_j \kappa_{ij} X_j^j \left(X_j^j\right)^T + \left(\Phi - \frac{1}{2}\left(\rho - \gamma^2\right) I\right)^T X^i + X^i \left(\Phi - \frac{1}{2}\left(\rho - \gamma^2\right) I\right)$$

  - $X_i^i \equiv$ the $i$ th column of $X^i$
  - $\Phi \equiv \Omega + \mu^2 \begin{bmatrix} X_1^1 & \cdots & X_n^n \end{bmatrix}^T$
- Algorithm: Given $\begin{bmatrix} X_\tau^1 & \cdots & X_\tau^n \end{bmatrix}$, update $\begin{bmatrix} X_{\tau-\Delta}^1 & \cdots & X_{\tau-\Delta}^n \end{bmatrix}$ by

$$-\frac{X_\tau^i - X_{\tau-\Delta}^i}{\Delta} = Q^i - \mu^2 \sum_j \kappa_{ij} X_{j,\tau}^j \left(X_{j,\tau}^j\right)^T + \left(\Phi_\tau - \frac{1}{2}\left(\rho - \gamma^2\right) I\right)^T X_\tau^i + X_\tau^i \left(\Phi_\tau - \frac{1}{2}\left(\rho - \gamma^2\right) I\right)$$

Back

# Summary of Equilibrium

| Description | Expression |
|---|---|
| Production strategy | $\mathbf{q}_t = N\mathbf{z}_t$ |
| R&D strategy | $\mathbf{x}_t = \mu \tilde{X} \mathbf{z}_t$ |
| Law of motion | $d\mathbf{z}_t = (\Omega \mathbf{z}_t + \mu \mathbf{x}_t)\, dt + \gamma \mathbf{z}_t dW_t$ |
| Profit of final producers | $\Pi_t^F / P_t = \mathbf{q}_t^T \left(\frac{1}{2}\Sigma\right) \mathbf{q}_t$ |
| Total operating profit of firms | $\Pi_t / P_t = \mathbf{q}_t^T \left(\frac{1}{2}\Sigma \circ \left(K + K^T\right)\right) \mathbf{q}_t$ |
| Labor income | $w_t L / P_t = \mathbf{q}_t^T \left(\frac{\zeta}{L} J\right) \mathbf{q}_t$ |
| Output | $Y_t = \mathbf{q}_t^T \left(\frac{\zeta}{L} J + \frac{1}{2}\Sigma + \frac{1}{2}\Sigma \circ \left(K + K^T\right)\right) \mathbf{q}_t$ |
| Consumption | $C_t = Y_t - \mathbf{x}_t^T \mathbf{x}_t$ |

Back

# Output and Expected Utility

- Output: $Y_t = \mathbf{q}_t^T Q \mathbf{q}_t$ where

$$Q = \frac{\zeta}{L} J + \frac{1}{2}\Sigma + \frac{1}{2}\Sigma \circ \left(K + K^T\right)$$

- Expected utility:

$$V(\mathbf{z}_t) \equiv \mathbb{E}_t \left[ \int_t^{\infty} \exp(-\rho s) C_s ds \middle| \mathbf{z}_t \right] = \mathbf{z}_t^T X \mathbf{z}_t$$

where $X$ is the solution of the Lyapunov equation (obtained from households' HJB equation):

$$0 = Q - \mu^2 \tilde{X}^T \tilde{X} + X \left( \Phi - \frac{1}{2}\left(\rho - \gamma^2\right) I \right) + \left( \Phi - \frac{1}{2}\left(\rho - \gamma^2\right) I \right)^T X$$

Back

# Social Optimum

- Static optimal allocation: $\mathbf{q}_t^* = N^* \mathbf{z}_t$ where $N^* \equiv \left\{ 2\frac{\zeta}{L} J + \Sigma \right\}^{-1}$
- Optimal output: $Y_t^* = \mathbf{z}_t^T Q^* \mathbf{z}_t$ where $Q^* = \frac{1}{2} N^*$
- Optimal expected utility:

$$V^*(\mathbf{z}_t) \equiv \mathbb{E}_t \left[ \int_t^\infty \exp(-\rho s) C_s ds \middle| \mathbf{z}_t \right] = \mathbf{z}_t^T X^* \mathbf{z}_t,$$

where $X^*$ is the solution of the Riccati equation (obtained from planner's HJB equation):

$$0 = Q^* - \mu^2 (X^*)^2 + X^* \left( \Phi^* - \frac{1}{2} (\rho - \gamma^2) I \right) + \left( \Phi^* - \frac{1}{2} (\rho - \gamma^2) I \right) X^*$$

- Optimal R&D: $\mathbf{x}_t^* = \mu X^* \mathbf{z}_t$
- Optimal technology transition matrix: $\Phi^* = \Omega + \mu^2 X^*$

Back

# Stochastic Process of Output

- Applying Itô's lemma,

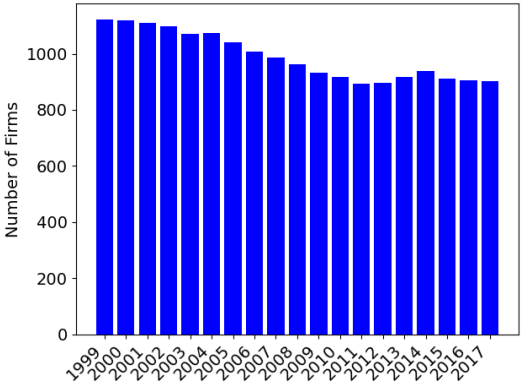
$$d\log Y_t = \left[\frac{\mathbf{z}_t^T\left(Q\Phi + \Phi^T Q\right)\mathbf{z}_t}{Y_t} + \gamma^2\left\{\frac{\sum_i z_{i,t}^2 Q_{ii}}{Y_t} - \frac{2\mathbf{z}_t^T Q\,\mathrm{diag}\left(\mathbf{z}_t^2\right)Q\mathbf{z}_t}{Y_t^2}\right\}\right]dt + \frac{2\gamma\mathbf{z}_t^T Q\,\mathrm{diag}\left(\mathbf{z}_t\right)}{Y_t}dW_t$$

where $Y_t = \mathbf{z}_t^T Q\mathbf{z}_t$ and $\Phi = \Omega + \mu^2\widetilde{X}$

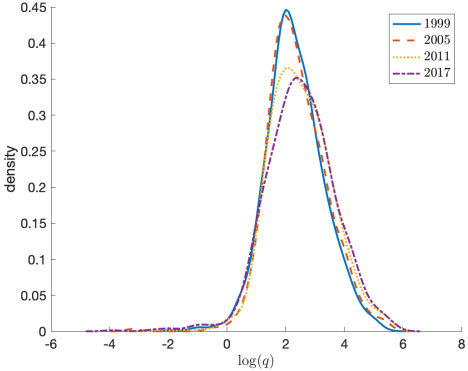
| Description | Expression |
|---|---|
| Tech Spillover | $\mathbf{z}_t^T\left(Q\Omega + \Omega Q\right)\mathbf{z}_t / Y_t$ |
| R&D | $\mu^2\mathbf{z}_t^T\left(Q\widetilde{X} + \widetilde{X}^T Q\right)\mathbf{z}_t / Y_t$ |
| Ito | $\gamma^2\left\{\sum_i z_{i,t}^2 Q_{ii}/Y_t - 2\mathbf{z}_t^T Q\mathrm{diag}\left(\mathbf{z}_t^2\right)Q\mathbf{z}_t/Y_t^2\right\}$ |
| Total | $\mathbb{E}\left[d\log Y_t\right]$ |

Back

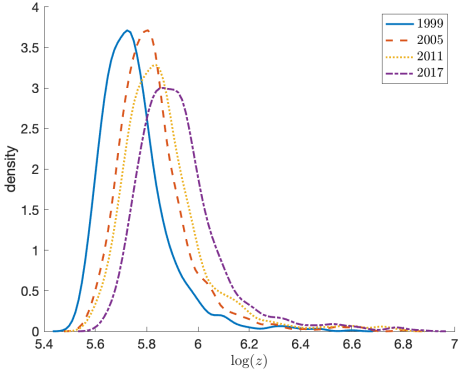
# Number of Sample Firms



# Distributions of Estimated Knowledge Capital and Quantity




(a) Log of Quantity




(b) Log of Knowledge Capital

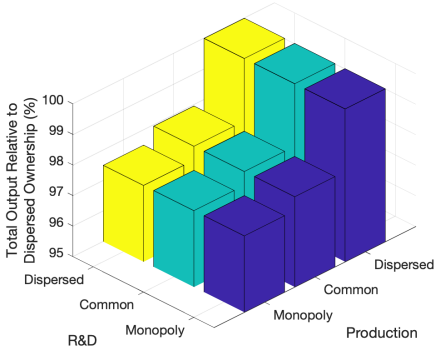
# First Stage

Back

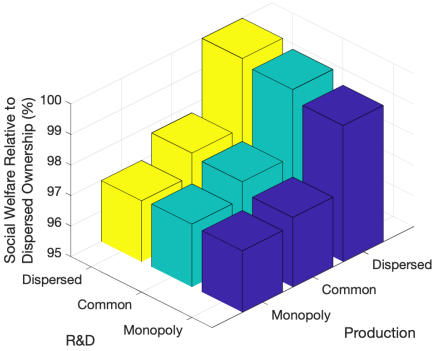
| | (1) |
|---|---|
| Dependent Variable: | $z_{i,t}$ |
| User cost of R&D | $-39.495^{***}$ |
| | (4.7044) |
| Year Fixed Effects | ✓ |
| No. observations | 12,947 |

SEs clustered by years and 4-digit naics industries are reported in parentheses.

# Different Ownership Structure: Output and Social Welfare



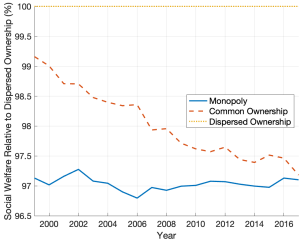

(a) Total Output



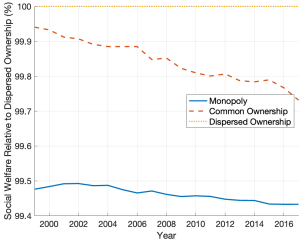

(b) Expected Social Welfare

- Ownership concentration in both R&D and production tends to reduce welfare Back

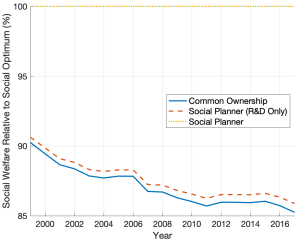
# Trends of Social Welfare




(a) Different Ownership

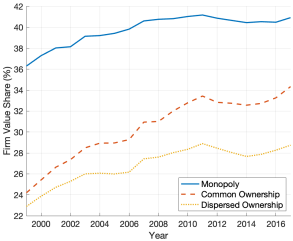



(b) Different Ownership (R&D Only)

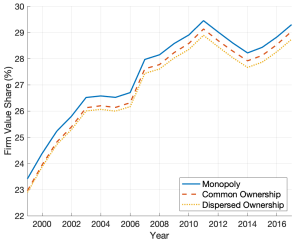



(c) Comparison with Social Optimum

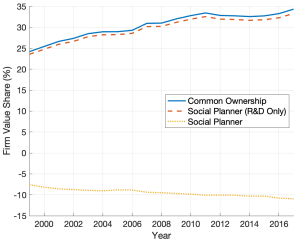
# Trends of Firm Value Share




(a) Different Ownership




(b) Different Ownership (R&D Only)




(c) Comparison with Social Optimum

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